

Wave 5.2R Stage 8 Weak Forward Compliance Priors Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 8 is complete as a valid negative result.

All 10 / 10 first-screen runs completed without failure. No weak-compliance candidate passed the complete multi-index gate, so the conditional stability continuation was correctly skipped.

The training-only diagnostic strongly supports a positive *population* association between applied torque and curve-mean TE: all 512 / 512 bootstraps are positive, with a 95% interval from $3.623920841e-05$ to $3.713770981e-05$ deg/Nm . The shuffled-torque control returns 50.39% positive support, as expected under loss of specificity.

That valid observable relationship does not transfer into useful local physics guidance. No weak-prior arm beats data-only C00, and all weak arms retain negative model-local derivatives for 35.1% to 44.3% of test conditions. H01 enforces a positive derivative everywhere, but materially underfits raw and mean TE.

Stage 5 H04 remains the qualified structured component entering Stage 9. No Stage 8 model replaces the accepted periodic GRU or becomes a production candidate.

Scope And Integrity

- dataset: polished_dataset ;
- input contract: setpoints only;
- surface: Fw ;
- accepted curves: 966 ;
- split: 675 train, 194 validation, 97 test;
- angular grid: 2048 uniform points;
- split signature:
c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16 ;
- first-screen seed: 314159 ;
- completed runs: 10 / 10 ;
- failed runs: 0 ;
- target-derived runtime inputs: none;
- validation or test targets used for prior estimation: no;
- official TE Curve Verification Pipeline: not run.

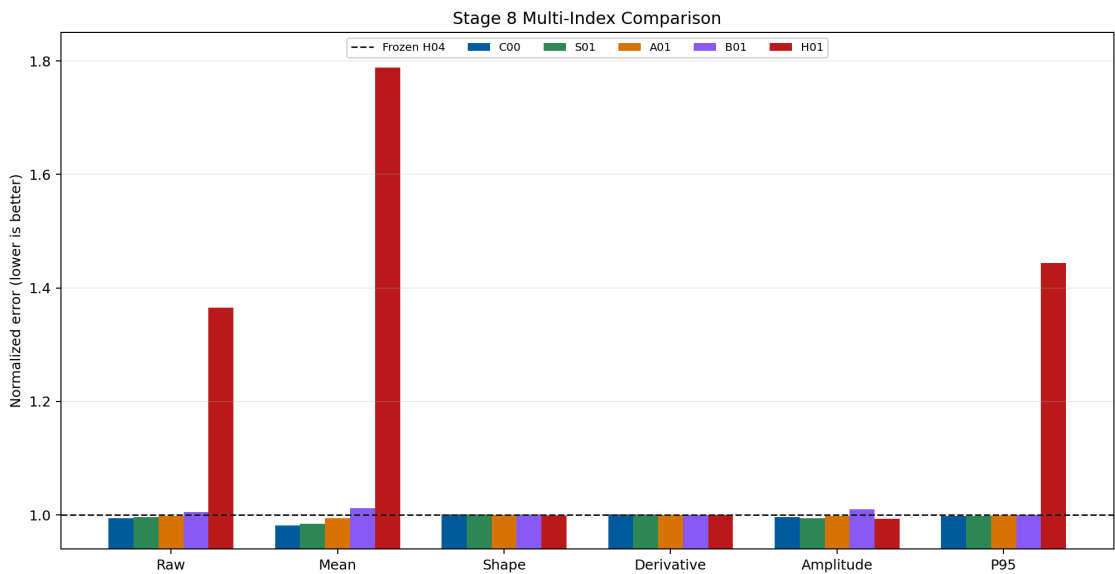
This stage intentionally tests an observable response prior, not an identified contact-stiffness law. The dataset does not contain ordered load-unload cycles, clearances, contact forces, hysteretic internal states, or component-level stiffness measurements required by the Xu mechanical formulation.

Candidate Matrix

ID	Formulation	Role
D00	frozen H04	immutable structured baseline
C00	data-only H04 fine-tune	learned control
S01	sign-only derivative penalty	weakest physics arm
B01	broad bootstrap interval	interval arm
W01	confidence-weighted interval	support-aware arm
T01	temperature-stratified interval	conditional arm
A01	delayed interval activation	curriculum arm
R01	adaptive interval weighting	optimization arm
N01	shuffled-torque interval	specificity control
H01	fixed compliance equation	misspecification control

Primary Results

ID	Raw [deg]	Mean [deg]	Shape [deg]	dTE/dT [deg/Nm]	Negative [%]
D00	0.0017259	0.0008844	0.0013555	-1.020e-08	60.8
C00	0.0017169	0.0008681	0.0013577	9.982e-08	37.1
S01	0.0017196	0.0008710	0.0013577	1.093e-07	37.1
A01	0.0017224	0.0008790	0.0013561	2.069e-08	44.3
B01	0.0017352	0.0008954	0.0013567	4.450e-08	35.1
N01	0.0017169	0.0008681	0.0013577	9.982e-08	37.1
H01	0.0023556	0.0015811	0.0013548	3.558e-05	0.0

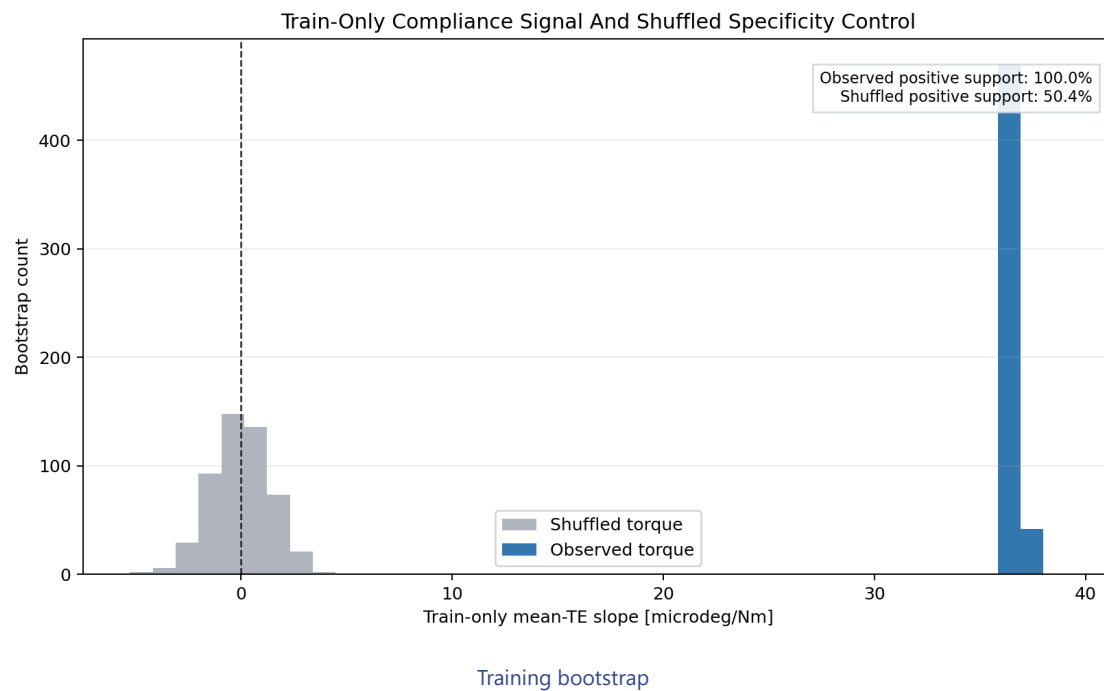


Stage 8 multi-index comparison

C00 is the raw-error leader at 0.001716862 deg . It improves raw MAE by 0.52% and mean MAE by 1.85% relative to frozen H04, while centered-shape MAE changes by -0.16% . This gain is attributable to bounded data-only fine-tuning, not to a compliance constraint.

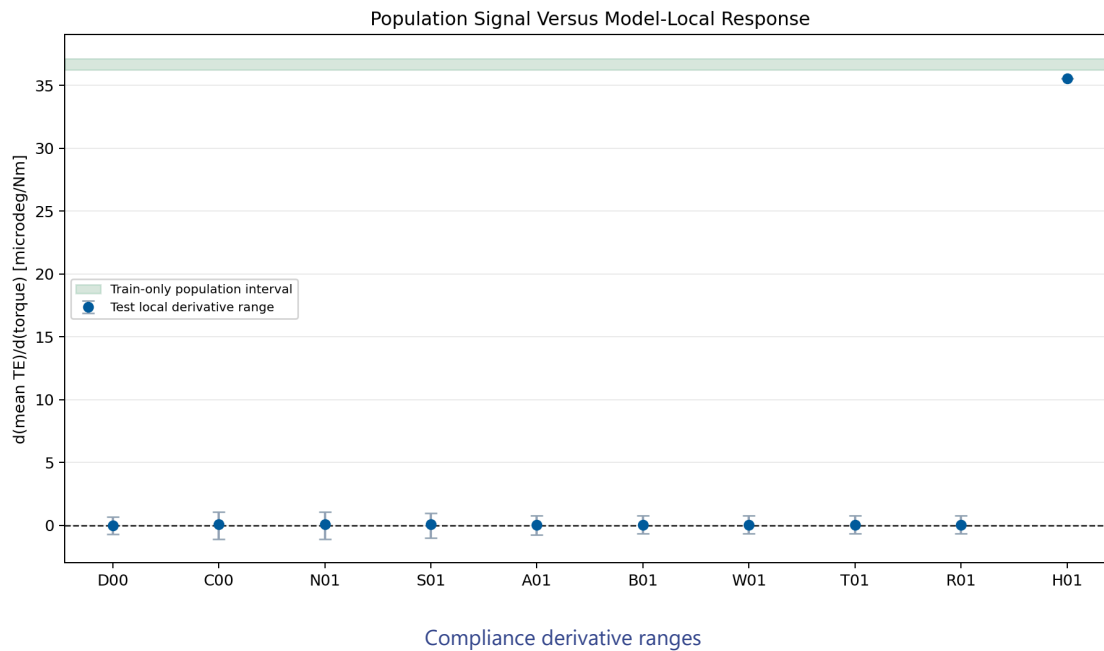
S01 improves raw MAE by 0.36% and mean MAE by 1.51% relative to frozen H04, but it is worse than C00 on both quantities and does not establish a consistently positive local response.

Train-Only Identifiability Diagnostic



The population slope is stable and specific to the real torque ordering. Its median corresponds to an effective response scale of 27282.7 Nm/deg. This number is a descriptive reciprocal slope, not an identified reducer stiffness. It conflates operating-condition sampling, offsets, temperature, speed, contact state, and unobserved hysteresis.

Local Derivative Behavior

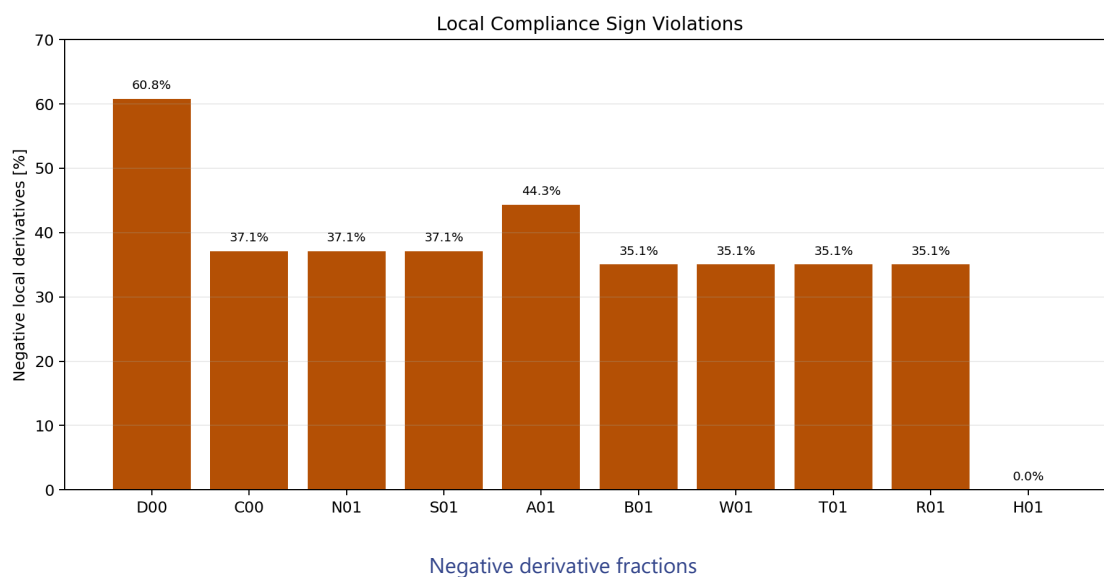


The green band is the train-only population interval. The learned model-local derivatives are roughly three orders of magnitude smaller and cross zero for every weak arm. The diagnostic therefore distinguishes two different claims:

1. higher-torque operating cells have higher mean TE on average;
2. each learned curve should respond monotonically to an infinitesimal torque

perturbation while every other input remains fixed. The first claim is supported. The second is not identified by this dataset and is not recovered by the tested penalties.

Sign Violations



H01 is the only formulation with zero sign violations. Its derivative is fixed near the train-only population slope, but raw MAE rises to 0.0023556 deg and mean MAE to 0.0015811 deg. This is direct evidence that the hard equation is misspecified for pointwise prediction.

Gate Matrix

ID	R-H04	R-C00	M-H04	M-C00	Shape	Curve	N01	Sign	Final
S01	pass	fail	pass	fail	pass	pass	fail	fail	fail
B01	fail	fail	fail	fail	pass	fail	fail	fail	fail
W01	fail	fail	fail	fail	pass	fail	fail	fail	fail
T01	fail	fail	fail	fail	pass	fail	fail	fail	fail
A01	pass	fail	pass	fail	pass	pass	fail	fail	fail
R01	fail	fail	fail	fail	pass	fail	fail	fail	fail

No candidate passes. S01 and A01 beat frozen H04 on raw and mean error, but neither beats C00 or the shuffled-control requirement, and neither has positive local derivatives throughout the test surface. B01, W01, T01, and R01 additionally regress raw and mean behavior.

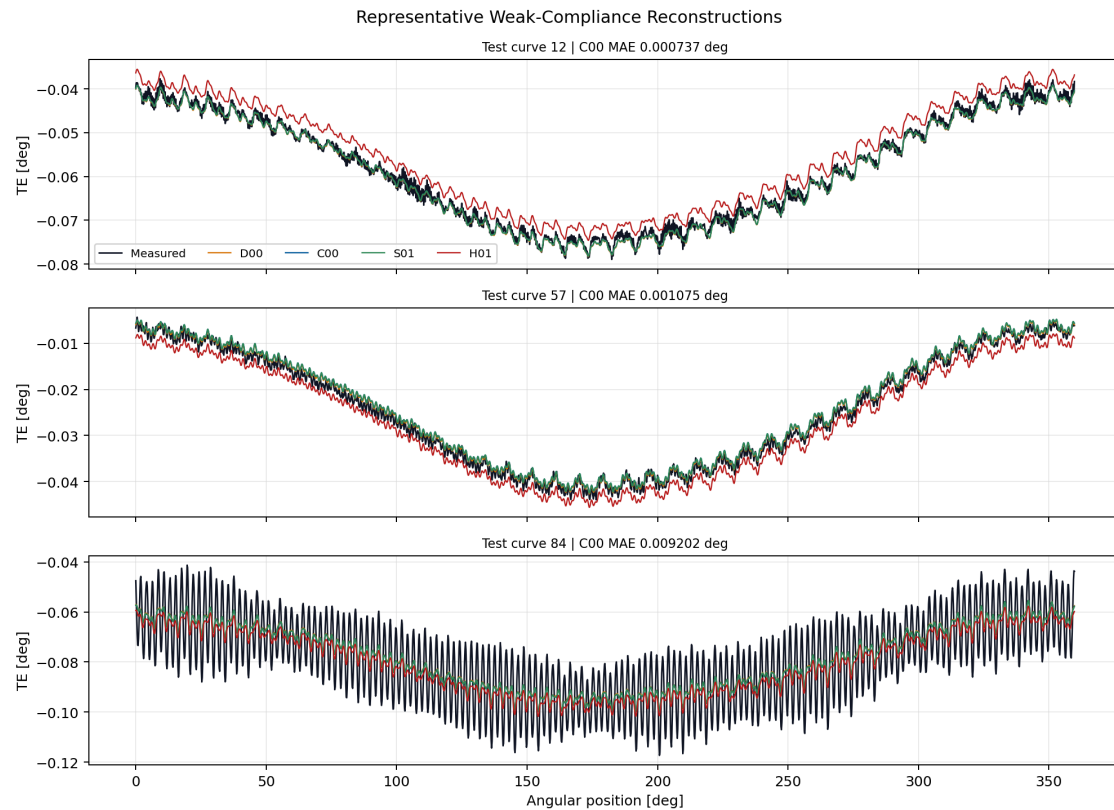
What Worked

- the train-only bootstrap found a strong positive population association;
- the shuffled-torque bootstrap removed that directional support;
- autograd derivatives were computed without target-derived runtime inputs;
- the weak-to-hard ladder exposed the tradeoff between predictive fit and derivative enforcement;
- all curve-first diagnostics and failure controls completed deterministically.

What Did Not Work

- no weak physics arm outperformed data-only C00;
- N01 is numerically identical to C00 because its shuffled interval does not activate a useful constraint;
- sign-only and delayed penalties leave 37.1% and 44.3% negative local derivatives, respectively;
- broad, weighted, temperature, and adaptive intervals converge to nearly the same inferior solution;
- the hard compliance equation removes sign violations only by sacrificing raw and mean accuracy;
- no candidate earns stability continuation or promotion.

Representative Full Curves



Representative curves

C00 and S01 remain visually close to frozen H04. H01 shifts whole-curve levels because the fixed compliance term dominates the mean response; its slightly better centered-shape metric does not compensate for the offset error.

Scientific Interpretation

The Stage 8 result does not show that compliance is absent. It shows that a cross-sectional population trend is insufficient to define a causal, pointwise constitutive residual. A PINN can compensate for an incomplete mechanical model only when the residual constrains the intended state without forcing the network to absorb systematic misspecification.

Here the missing load history, direction reversals, contact regime, clearance, and internal torsional states make the local derivative ambiguous. The network therefore treats the compliance penalties as optimization bias rather than additional identifiable information.

This evidence justifies moving to Stage 9, where causal history is tested as the missing information channel through temporal analytical-residual models. Stage 9 must preserve the same negative-control and data-only comparison discipline.

Program Decision

- Stage 8 status: complete, valid negative result;
- completed runs: 10 / 10 ;
- stability runs: 0 , correctly skipped;
- raw-error leader: C00 data-only control;
- promoted Stage 8 candidate: none;
- retained component: Stage 5 H04;
- production or registry promotion: no;
- next step: Stage 9, Temporal Analytical-Residual Models.

Artifact Map

- campaign:
output/training_campaigns/2026-07-29-18-19-20_wave52r_stage8_weak_forward_compliance_priors_2026_07_29/ ;
- gate summary:
output/analysis/wave_5_2r/stage8_weak_forward_compliance_priors/closeout/stage8_exit_gate_summary.yaml ;
- bootstrap:
output/analysis/wave_5_2r/stage8_weak_forward_compliance_priors/stage8_training_only_bootstrap.yaml ;
- preflight:
output/analysis/wave_5_2r/stage8_weak_forward_compliance_priors/stage8_preflight_validation_summary.yaml ;
- C00 checkpoint:
output/training_runs/weak_forward_compliance_priors/2026-07-29-18-19-22__stage8_c00/best_model.pt ;
- model report:
doc/reports/analysis/model_development_waves/wave_5_2/physics_guided_pinn_reassessment/[2026-07-29]/stage8_weak_forward_compliance_priors/stage8_weak_forward_compliance_priors_model_report.md .